

Project Report: Predicting NBA Game Outcomes by Understanding Quarter-to-Quarter Performance Trajectories

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Abstract

Current forecasting approaches on NBA game outcomes rely on aggregate or pre-game statistics, overlooking how within-game dynamics evolve. This project will examine to what extent quarter-level performance dynamics predict final game outcomes. Using the `nba_api` package and the `playbyplayv3` endpoint, we assembled a dataset of 3,463 games and 1,697,676 play-by-play events across three NBA seasons (2022-23 through 2024-25). Data was chronologically partitioned into training, validation, and test sets. Exploratory analysis using a linear model showed that cumulative score differential at the end of Q3 was the dominant predictor, with non-scoring features contributing negligible incremental R^2 . We then evaluated three models of increasing complexity: a logistic regression baseline, tree-based ensembles (Random Forest and XGBoost), and a time-aware GRU. XGBoost achieved the best performance with a log-loss of 0.3254 and F1-score of 0.9361, while the GRU showed limited gains given the short three-timestep sequences. These results suggest that quarter-level dynamics carry meaningful predictive signal, but cumulative score differential captures most of it, leaving nonlinear interactions as the primary remaining source of gain.

Introduction

Predicting NBA game outcomes is commonly based on aggregate or pre-game statistics, but these approaches do not capture how performance evolves during a game. Quarter-level dynamics provide a more detailed view of how teams adjust over time and how momentum shifts impact results. Understanding these patterns can improve prediction accuracy and offer insight into in-game decision making.

In this project, we model game outcomes using quarter-by-quarter performance trajectories derived from play-by-play data. The input to our models consists of time-ordered game data aggregated into features such as cumulative score differentials and other derived metrics. The output is a binary prediction of whether a team wins or loses a game. We evaluate three models: a logistic regression baseline, an ensemble model, and a time-aware model designed to capture temporal structure.

Our approach is motivated by the observation that cumulative score differential is strongly correlated with game outcomes but not fully predictive, especially in close games. This suggests that additional structure, such as nonlinear relationships and temporal patterns, may improve performance. We compare models of increasing complexity to assess how much predictive value can be gained beyond simple scoring-based features.

Related Work

Benz et al. [1] investigated how momentum influences shot success in the NBA, defining negative momentum as a team being outscored over the previous five minutes. Their analysis revealed substantial drops in shooting accuracy under high-pressure situations: two-point field goal percentage fell from 47.5% to 37.2%, three-point percentage dropped from 35.3% to 22.0%, and free throw accuracy declined

from 76.0% to 71.4%. This work demonstrates that within-game dynamics meaningfully affect player performance at fine temporal resolution. However, its scope is limited to shot-level outcomes rather than full-game prediction, leaving open the question of whether these momentum effects translate into measurable gains in win-loss forecasting. Perricone, Shaw, and Swiechowicz [2] addressed the game-prediction task directly, using aggregate team statistics and recent-game features from the NBA API. Their best benchmark reached approximately 74% accuracy, with an SVM model achieving 67.6% and logistic regression reaching 67.1% on the test set. While their feature set incorporated some quarterly information, the granularity remained coarse with reliance on aggregate trends rather than within-game progressions.

These works highlight a gap in our project targets: Benz et al. show that momentum matters at the shot level but stop short of game prediction, while Perricone et al. predict outcomes but underutilize within-game temporal structure. By modeling quarter-by-quarter performance trajectories, we examine whether intermediate-resolution features can improve outcome prediction beyond aggregate baselines.

Dataset

Data Source and Scope: We constructed our dataset using the `nba_api` Python package, drawing from the `playbyplayv3` endpoint to capture three full NBA seasons: 2022–23, 2023–24, and 2024–25. From this source, we first pulled a master list of 3,463 games using their unique `game_id` identifiers. We then queried each game's full play-by-play log to build a composite event-level dataset containing 1,697,676 individual actions made by both home and away teams. The prediction target throughout this study is binary: whether a given team wins or loses, conditioned on its quarter-by-quarter performance trajectory.

Preprocessing and Feature Construction: Several preprocessing steps were required to convert raw play-by-play events into model-ready features. To consistently align team metadata across records, we produced two mapping tables. First we built a `team_map` to standardize team identifiers and a `game_map` to store game-level information including home and away designations. Since free throws did not consistently contain a `pointValue` field in the raw data, free throws were assigned a value of 1 point to ensure accurate scoring aggregation. For finer-grain quarter analysis, rather than relying directly on the endpoint's cumulative `homeScore` and `awayScore` fields, we wrote a function to extract rows where `actionType == "period"` and `subType == "end"`, allowing us to derive quarter scoring and cumulative point differentials at well-defined break points. We did not apply imputation for missing values: inspection revealed that nulls in the raw data did not represent missing observations but rather encoded inapplicability (e.g., `teamId` was null for events affecting both teams such as replay reviews), and treating them as missing would have introduced spurious signal.

Data Splits: To support time-aware modeling and avoid temporal leakage, we used a chronological train/validation/test split rather than random partitioning. The training set comprises the full 2022–23 ($n = 1,145$) and 2023–24 ($n = 1,091$) seasons, totaling 2,236 games. The 2024–25 season was split 50/50 into validation ($n = 521$) and test ($n = 522$) sets in chronological order. Overtime games were excluded across all splits to maintain homogeneous game durations and reduce variance introduced by extended play, since our quarter-level features are designed around a fixed four-quarter structure. This chronological partitioning simulates realistic deployment conditions, where a model trained on prior-season data must generalize to a future, unseen season.

Exploratory Data Analysis: Analysis was conducted to understand the statistical and temporal drivers of game outcomes utilizing features from the first three quarters. Both outcome-proximal variables such as cumulative score differentials and explanatory variables such as shot efficiency, measures of momentum (e.g. rates of change in score differentials), and possession metrics were feature engineered from the playbyplay dataset. Initial segment-based analysis indicated that winners and losers had nearly identical shot distance profiles as seen in **fig. 1**. Non-scoring feature differentials were significantly different across winners and losers (**fig. 2**) where rebounds ($r=0.218$, $p<0.001$) and fouls ($r=0.097$, $p<0.001$) were more correlated with the final score margin for winning a game. In contrast, turnovers ($r=-0.155$, $p<0.001$) and free throws ($r=-0.137$, $p<0.001$) were negatively correlated with winning score margins. Although significantly significant, the correlation coefficients were modest at best and this was further supported when a baseline linear regression model using cumulative difference in points by Q3 (`cum_diff_Q3`) did not improve in R^2 with additional action/shot types as predictors as seen in **Table A1**. Subsequent modelling strategies were heavily informed by the lack of predictive strength from shot distance, momentum, and action type.

Methods

Logistic regression was utilized as the baseline model inspired by Perricone, Shaw, and Swiechowiec [2] utilizing `cum_diff_Q3` as the single independent variable. As other variables were revealed to have little effect on increasing predictive power of the linear regression model during EDA (**Table A1**), `cum_diff_Q3` was selected as the predictor. The baseline model was built with Keras Sequential API as a single-layer neural network using sigmoid activation to predict the binary outcome of a home team win or an away team win. Binary cross entropy was then chosen as the loss function and Adam as the optimizer.

To capture the nonlinear interactions and complex temporal dynamics that simple linear baselines (such as Logistic Regression) might fail to recognize, we introduced tree-based ensemble methods—specifically Random Forest and XGBoost. The primary motivation for generating these ensemble models is to leverage quarter-by-quarter scoring trajectories to identify underlying momentum shifts. For example, a team struggling in the first quarter but surging in the third quarter may present a compounding interaction effect. By simultaneously running an ensemble of independent or boosted decision trees, these methods can automatically capture these multi-layered feature interactions (e.g., the combined effect of home-court context and quarter-specific differentials) without the need to manually compute higher-dimensional polynomial expansions. Both models were trained on Q1-Q2 score differential features: cumulative scores (`cum_diff_Q`), non-cumulative score (`Q_diff`), opposing team points (`opp_Q_pts`), and if the team is playing at home (`is_home`).

A gated recurrent unit (GRU) model was selected as the third modeling approach to model temporal dynamics. GRUs are well-suited to model how game context dynamically evolves such as shifts in scoring rates, shooting efficiency, turnovers, and rebounds throughout the progression of a game. Under the hypothesis that teams adjust their strategies in response to game states and with mounting pressures of arriving at the end of the game, GRUs could elucidate temporal patterns and non-linear predictive relationships. The initial pre-tuned model was built with TensorFlow Keras by creating quarter-level timesteps from Q1 to Q3 in which five differentials were calculated: cumulative score (`cum_diff_Q`), non-cumulative points (`pts_diff_Q`), field goals (`fg_diff_Q`), three-point rates (`tpr_diff_Q`), and turnovers (`tov_diff_Q`) for team_A (anchor team) and team_B for each game. The 32 unit GRU layer would receive input tensors in the shape of `[batch_size, 3, 5]` followed by a dropout layer, dense layer with ReLu

activation, another dropout layer, and finally, a dense layer with sigmoid activation that outputs a team_A win.

Experiments, Results & Discussion

The logistic regression model improved meaningfully from validation to test across all metrics F1, MAE, and log loss indicating changes to learning rate and number of epochs yielded moderate gains without overfitting. The logistic regression model demonstrated surprisingly robust performance for a linear classifier. It achieved a log-loss of 0.3280, an F1-score of 0.8710, and a mean absolute error of (MAE) of 0.1379 on the test set. Weight analysis confirmed that the third quarter cumulative differentials served as the primary predictive driver, effectively integrating prior game momentum into a high-fidelity signal. This suggests a remarkably linear relationship between late-game scoring margins and final outcomes, mirroring findings by LaRow et al. [3] and Tsagris et al. [4]. While the model lacks non-linear interaction capabilities of ensemble frameworks like XGBoost, its strong performance relative to more complex architectures establishes an 87% accuracy threshold as a probable “predictive ceiling” for NBA game outcomes.

As shown in Table A2, both ensemble models exhibited strong generalizability, with improvements consistently from validation to test performance across all metrics. Most notably F1 scores of 0.9245 to 0.9361 for XGBoost and 0.9128 to 0.9332 for random forest. This suggests that the tuning enhanced model fit without overfitting to the validation set. When evaluating the final tuned model on the test dataset, the ensemble framework possessed exceptionally robust discriminatory capability, evidenced by both models scoring over 0.93 in ROC AUC (**fig. 3**). Upon a side-by-side comparison, the gradient-boosted trees framework (XGBoost) demonstrated a moderately stronger structural fit compared to Random Forest. XGBoost reported a notably lower, well-calibrated Log-Loss score (0.3254 opposed to 0.3373) and marginally better precision in estimating the point-spread (MAE of 6.63 points). This signifies that XGBoost's sequential boosting framework was slightly better equipped to adjust decision boundaries on statistically rigid edge cases found within close-game scenarios.

The initial GRU model fitted to the training data yielded comparable performance to the baseline across the both data sets where training achieved an accuracy of 0.8399 and validation an accuracy of 0.8407. Based on the loss and accuracy curves of the training and validation (**fig. 4**), the model indicated a high level of generalizability where it was capable of applying the learned temporal dependencies and patterns arising from the differential features within the training set to the validation set. There is a slight decrease in accuracy observed with the validation dataset however, this is most likely attributable to the inherent variability and noise in intergame dynamics as well as the smaller sample size. Using grid search to hyperparameter tune across learning rate, batch size, number of epochs, dropout rate, hidden units, and dense units, the final tuned model yielded marginal improvements in performance predicting on the test dataset as seen in **Table A2**. The tuned GRU performed the worst on the test set relative to the other models, with the highest test log loss (0.3646) and lowest test F1 score (0.8332). It also indicated a higher error in regression terms with a test MAE of 0.2203 indicating the GRU model had the poorest classification performance, probability calibration, and widest spread from the true outcomes.

Logistic regression served as a strong baseline which aligned with the findings of LaRow *et al.* [3] which found that logistic regression outperformed their SVM when predicting scoring streaks, attributing the

relatively linear relationship between key game state features such as scoring differentials and win/lose outcomes. The `cum_diff_Q3` feature explained the vast majority of predictive variance with all additional features which was also indicated as the most important feature within the XGBoost model based on ablation studies. This observation is consistent with Tsagris *et al.* [4], who found that half-time point differential was the single most important predictor of NBA game-level outcomes across three seasons. The ensemble models meaningfully outperformed the baseline indicating that for edge case games, it could capture quarter-level dynamics and residual signals that linear models cannot isolate. Chen *et al.* [5] reached an analogous conclusion in their momentum analysis where their elastic net model's ability to model interactions among scoring and environment variables revealed predictive relationships. Although the GRU's model was thought to have underperformed due to the large quarterly timesteps, when an alternative model was evaluated using more fine-grained 10-event time steps, performance remained largely unchanged with an accuracy of 0.8426, log loss of 0.3622, and f1 of 0.852. Although GRUs are meant to learn long-range sequential dependencies, Qiu et al. [6] demonstrated, the relationship between momentum and game outcomes diminishes monotonically from the first to the fourth quarter, meaning that earlier game states do not carry forward hidden information that later states fail to encode.

Conclusions (1–2 paragraphs)

Holistically, XGBoost yielded itself to be the best performing model achieving strong performance across classification and probability-based metrics. While ablation studies indicated that the model relied heavily on `cum_diff_Q3` similarly to logistic regression from a functional standpoint, XGBoost is capable of partitioning the feature space to isolate non-linear interactions and refine predictions through sequential boosting. From a practical perspective, the results of our project have constrained real-world usefulness in the context of informing game-play and coaching strategies. Since both the baseline and highest performing models have a disproportionate reliance on `cum_diff_Q3`, it largely confirms a non-informative intuition that teams leading during later game states are more likely to win. As shown in **fig. 5**, the win and loss distribution of `cum_diff_Q3` overlap substantially within the ± 10 point range and where the GRU's predicted probabilities in this region scatter considerably. To address this, it is recommended that the next steps would be to reframe the problem towards more actionable signals such as predicting outcomes earlier in the game and incorporating longer season-wide dynamics. Additionally, there is potential to focus specifically on close games that are roughly within plus or minus 10 points at the end of Q3 as these provide impactful insights where outcome prediction uncertainty is the highest.

Contributions

- Grace Suh - Authored GRU model notebook and contributed to methods, discussions, and conclusion sections.
- Kelly Yeh - Authored MODEL-logistic-regression.ipynb, EDA-features.ipynb, contributed to methods, discussions, and conclusion sections.
- Hailin Wang - Authored EDA_points.ipynb, contributed to Abstract, Introduction, Dataset sections. Wrote related works section
- Junyoung Yoon - Authored MODEL-ensembles.ipynb, contributed to methods, discussions, and conclusion sections.
- [GitHub Repository](#)

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Appendix

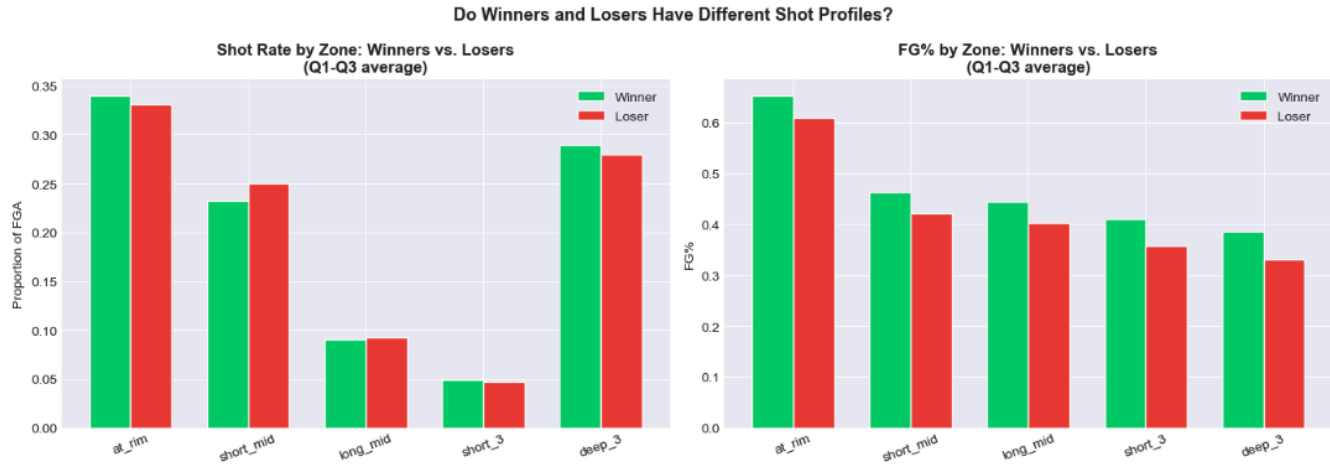


Fig 1. Comparison of shot distance profiles between winners and losers (Q1-Q3 averages). Both segments indicate highly similar proportions of shot rate and successful field goals across all shot zones with modest differences suggesting shot distance patterns may have minimal impact on game outcome.

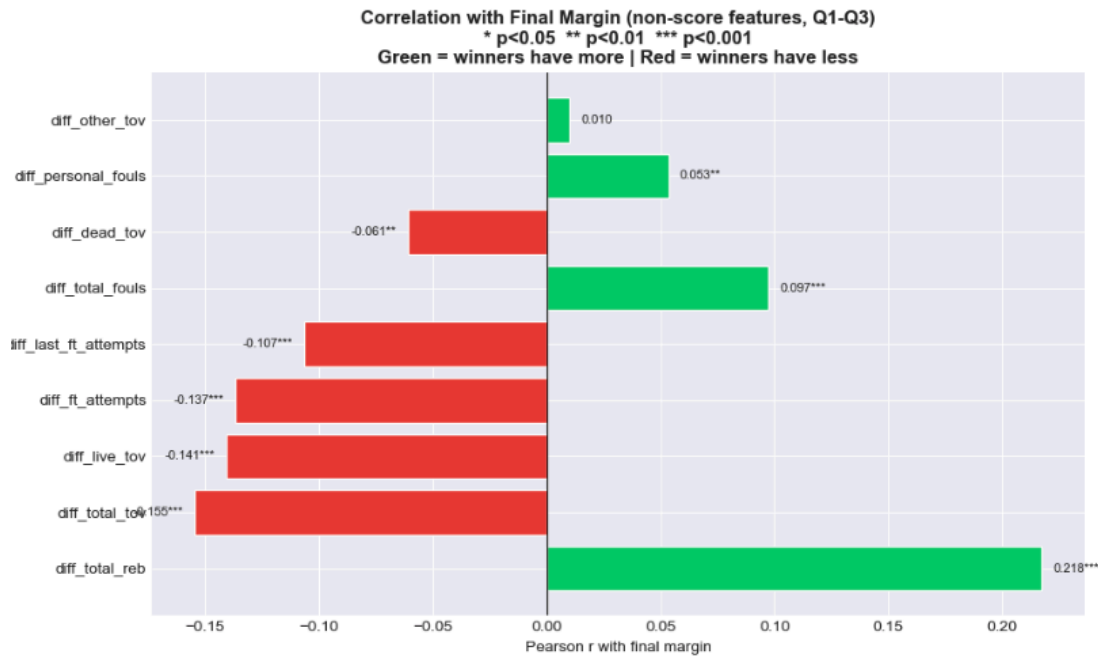


Fig. 2. Correlation between non-scoring feature differentials and final game margin using Q1-Q3 data where total rebounds exhibit the strongest positive association. Turnovers and free throw attempts are negatively associated with performance indicating modest importance of possession control.

Table A1. Incremental explanatory power of candidate features beyond cum_diff_Q3 estimated via linear regression during EDA. All features indicated marginal or negligible contribution to ΔR^2 suggesting they are either redundant, subsumed, or highly correlated by the cumulative scoring differential.

Category	Feature	Feature Description	ΔR^2
Baseline	cum_diff_Q3	Cumulative point differential at end of Q3	N/A
Momentum	Q3 Momentum	Change in scoring differential within Q3	+0.0008
Shot Zone	Zone Rates + FG%	Zone shot rates + zone field goal %, Q1-Q3	-0.0066
Non-Score	Rebounds	Total rebounds (offense + defense) per team, Q1-Q3	-0.0012
Non-Score	Fouls	Total fouls committed, Q1-Q3	-0.0004
Non-Score	FT Attempts + PCT	Free throw attempts + free throw make rate, Q1-Q3	-0.0007
Non-Score	Turnover Type	Live-ball (steals) vs dead-ball turnovers, Q1-Q3	-0.0003

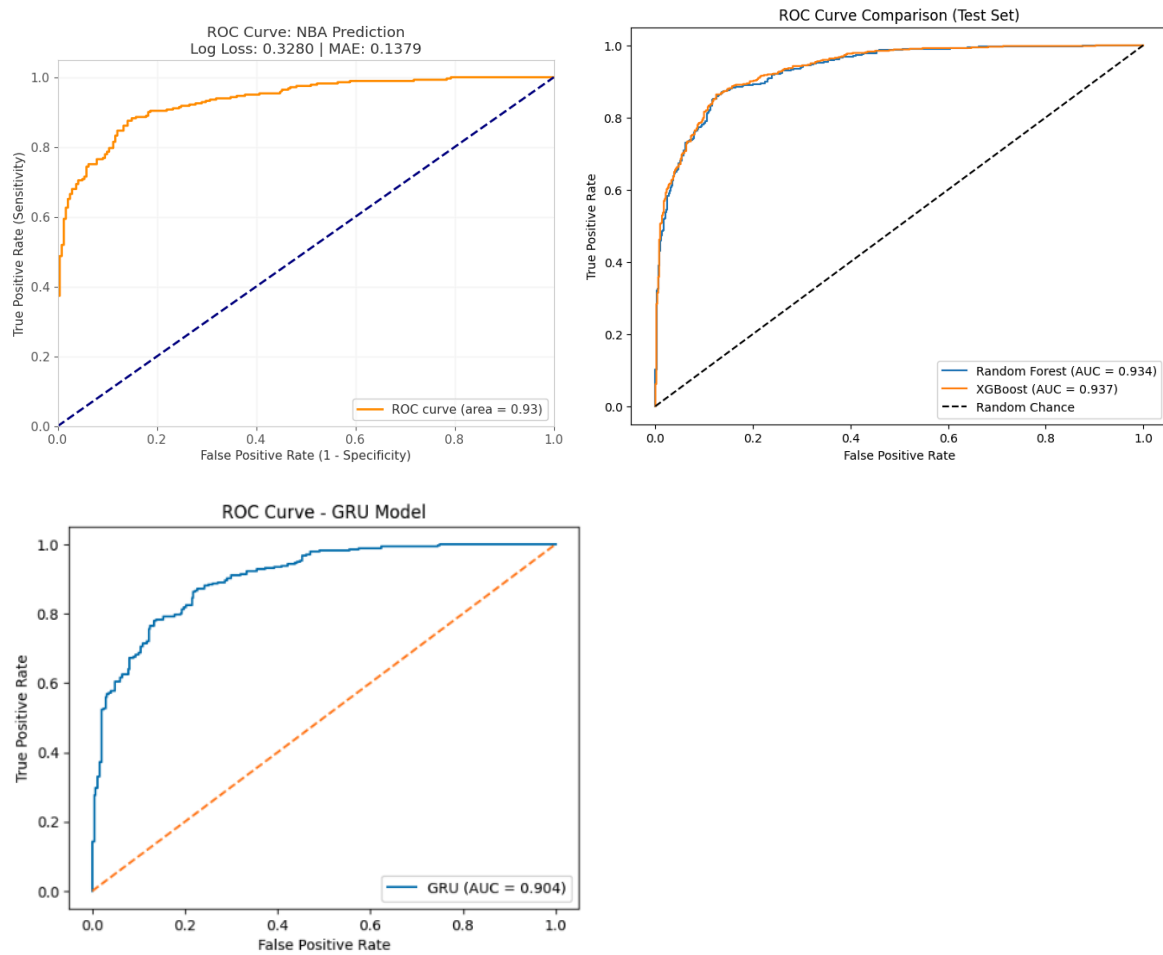


Fig. 3 ROC curve comparisons across models on the test set where the logistic regression baseline (top left) achieves high discriminative ability while ensemble models (top right) marginally outperform it with both XGBoost and random first attaining the highest AUC values. The GRU model (bottom) performs comparably but does not indicate substantial improvement over baseline.

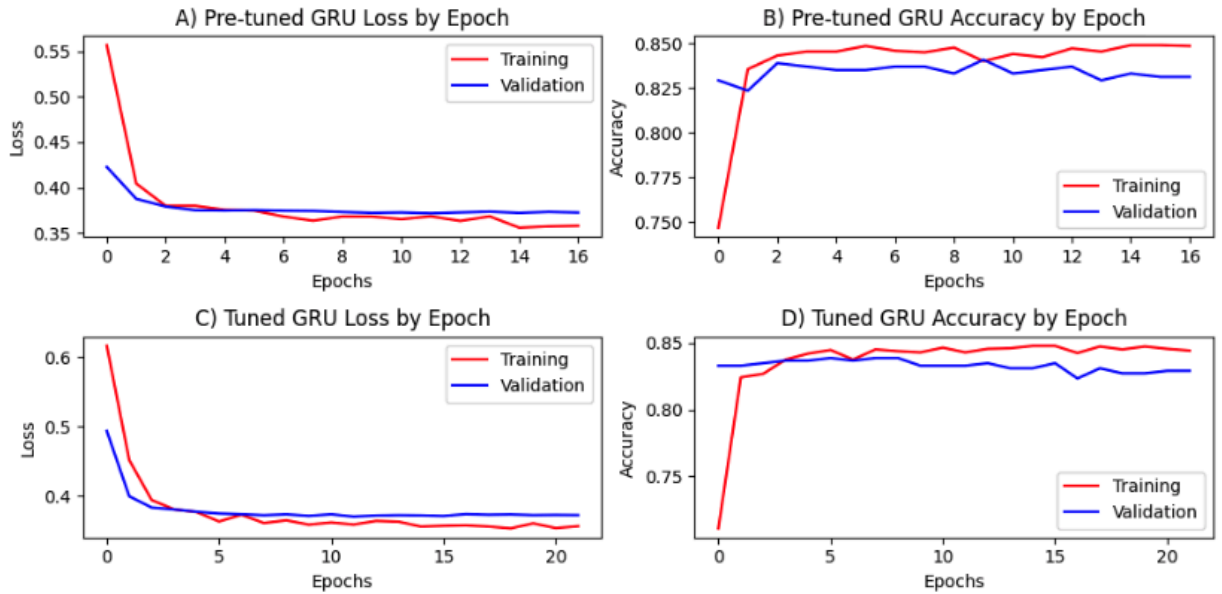


Fig. 4 Comparison of training and validation loss (A, C) and accuracy (B, D) across epochs for pre-tuned and tuned GRU models. Both configurations demonstrated rapid convergence and alignment between training and validation curves suggesting high generalization and lack of overfitting to the training data. The hyperparameter tuned model yields only modest performance gains.

Table A2. Performance comparison of pre-tuned models evaluated on the validation dataset (n=521) and hyperparameter-tuned models evaluated on the test dataset (n=522)

Model	Val Log Loss	Test Log Loss	Val F1	Test F1	Val MAE	Test MAE
Logistic Regression	0.3832	0.3280	0.8252	0.8710	0.1919	0.1379
Random Forest	0.3541	0.3373	0.9128	0.9332	0.1452	0.1322
XGBoost	0.3422	0.3254	0.9245	0.9361	0.1418	0.1331
GRU	0.3663	0.3646	0.8432	0.8333	0.2216	0.2203

Table A3. Final hyperparameter configurations for each model, selected based on optimal performance on the validation dataset and used for final evaluation on the test set.

Model	Final Hyperparameters
Logistic Regression	lr=0.01, batch_size=32, epochs=50
Random Forest	n_estimators=100, max_depth=6
XGBoost	lr=0.05, n_estimators=100, max_depth=4
GRU	lr=0.001, batch_size=64, epochs=20, dropout=0.2, hidden=32, dense=16

Fig. 5 Distribution of cum_diff_Q3 (left) and GRU predicted win probability as a function of cum_diff_Q3 (right). The left shows a substantial overlap in win and loss distributions within the ± 10 point change indicating high outcome uncertainty for competitive games. The probability distribution shows a sigmoidal relationship with considerable scatter within the 0.3-0.7 range.

